Department of Computing

Web Design and Development

Business Processes - Modelling & Design

Assignment: **Conceptualising the Problem Solution**
Due: **02nd March 2026**
Group: **THURSDAY GROUP 5**
Project Title: **PRIMARY STUDENT GRADING SYSTEM**

Instructions

a) This document acts as a reference or template to aid you in the design and development of web products based on the proposed problem ideas.
b) Fill in the respective sections with all necessary contents based on your group project ideas. Submit via Google Classroom as a soft copy and hand in a printed hard copy. A single submission is expected per group.
c) Any form of plagiarism will attract severe penalties. **All rules of academic writing apply**.
d) Any other document **format** other than what is contained and specified herein will **NOT** be assessed. This cover page must not overflow to the next page.

Group Members

#	Registration Number	Name (*as indicated on the portal*)	Score	Score	Score
1	BSC/INF/09/24	Mike Prosper Kamanga			
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Problem Definition

Primary school teachers in Malawi currently rely on manual methods to record, calculate, and distribute end-of-term student results. This approach is time-consuming, prone to arithmetic inaccuracies, lacks consistency across classes, and makes it difficult to store and retrieve records efficiently. There is no centralised platform for tracking student academic performance, generating standardised report cards, or enabling headteachers to access class-wide grade overviews.

Project Goal/Objective

To design and develop a web application that assists primary school teachers in grading end of term results for their students. The system will automate grade computation, produce standardised PDF report cards, support multi-role access (Administrator, and provide an overview of student academic performance.

Roles/Users

- *Administrator – manages teacher accounts, assigns classes and subjects*;
- Teacher – registers students, enters grades, verifies records, and generates report cards for their class.
- Head Teacher – views class and school-wide grade overviews and reviews generated reports.

Functional Requirements

List all functional requirements for your software (web API) application.

- Registration
  - **FR1.1** The system shall allow an administrator to create a teacher account.
  - **FR1.2** The system shall allow the administrator to assign the subject(s) and class that a teacher teaches.
  - **FR1.3** The system shall allow the teacher to register his/her students.

- Grade Processing
  - **FR2.1** The system shall compute the final end of term grades and generate a school report for the headteacher to view.
  - **FR2.2** The system shall generate report cards and grade records in pdf form.
  - **FR2.3** The system shall perform sort and filter operations on student grade records.
  - **FR2.4** The system shall enable the teacher to edit the grade of a student if a mistake was made.
  - **FR2.5** The system shall display the entered grades to the teacher for cross checking before processing.
  - **FR2.6** The system shall display an overview of student grades.

- Password Management
  - **FR3.1** The system shall allow the teacher to create or change their account password.

- **FR3.2** The system shall verify identity of the teacher when changing password through two step authentication.

- Authentication
  - **FR4.1** The system shall enable users to login and logout

REST Endpoints

Using a tabular format, list all possible endpoints for the software application being proposed. Each endpoint must include an associated description of the data it handles.

For example:

Module	Endpoint	Method	Description	Request Body / Parameters
Auth	/api/v1/auth/login	POST	Authenticate a user (teacher, headteacher, or administrator) and return a session token	{ username, password }
	/api/v1/auth/logout	POST	Logout user	–
	/api/v1/auth/me	GET	Retrieve the profile of the current authenticated user	–
	/api/v1/auth/reset	PUT	Initiate a password reset; sends a verification code to the user's registered email	{ email }
	/api/v1/auth/verify	POST	Comple twostep identity verification sent to the user's email	{ otp }
	/api/v1/auth/password	PUT	Set a new password after identity has been verified	{ newPassword, confirmPassword }
Users	/api/v1/users	POST	Create a new teacher/headteacher account in the system – by administrator only	{ name, email, ... }
	/api/v1/users	GET	List all teacher accounts	Query params
	/api/v1/users/:id	GET	Retrieve full profile details for a specific teacher/ user account by their ID	Path: id

	/api/v1/users/:id	PUT	Update account details for a specific teacher	{ ... }
	/api/v1/users/:id	DELETE	Permanently delete a teacher account – admin only	Path: id
	/api/v1/users/:id/subjects	GET	List all subjects assigned to a specific teacher	Path: id
	/api/v1/users/:id/class	GET	Cet the class that a specific teacher is assigned to	Path: id
	/api/v1/users/:id/subjects	POST	Assign one or more subjects to a teacher – admin only	{ Path:id, … }
	/api/v1/users/:id/class	PUT	Assign or change the class a teacher is responsible – admin only	{Path: id, .., ClassId }
Students	/api/v1/students	POST	Register a new student - authenticated teacher's only	{ fullName, dateOfBirth, class, ... }
	/api/v1/students	GET	List all students; teachers see only their class - supports filtering and sorting	Query Params
	/api/v1/students/:id	GET	Retrieve personal details and class information for a specific student	Path: id
	/api/v1/students/:id	PUT	Update details for a specific student record	{ Path: id, details..}
	/api/v1/students/:id	DELETE	Remove a student from the system - class teacher only	Path: id
	/api/v1/students/:id/grades	GET	Retrieve all grade records for a specific student across all subjects	Path: id
Reports	/api/v1/reports/generate	POST	Generate end-of-term report cards for all students in each class	{ class, term }
	/api/v1/reports/student/:id	GET	Download the PDF report card for a specific student for a given term	{ Path: id; term }}
	/api/v1/reports/class/:class	GET	Download a consolidated PDF grade record for an entire class	{ Path: class; term }

	/api/v1/reports/class/:class/overview	GET	Retrieve a summary report of class performance including averages and rankings	{ Path: class; term }